

Robustifying Profile Information Propagation in Profile-Guided Optimization

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Who Am I ?

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- Cyberchallenge.it 2024 finalist



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- Capture-the-Flag player at the official Sapienza CTF team



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- Cyberchallenge.it 2024 finalist
- Capture-the-Flag player at the official Sapienza CTF team
- Contributed to the LLVM compiler infrastructure by reporting undiscovered bugs



Performance is Critical

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- **Profile-Guided Optimization** is a way to achieve great performance
 - Optimization process **tailored** to the workload of the compiled program
 - **Profile** = Execution frequencies of program regions
 - Eliminates the need to infer such information
 - Perfect for high-load applications

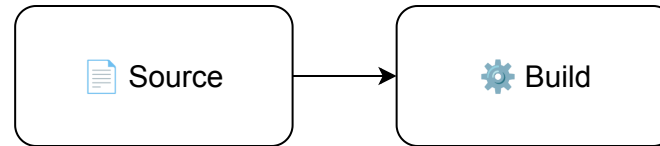
Profile-Guided Optimization

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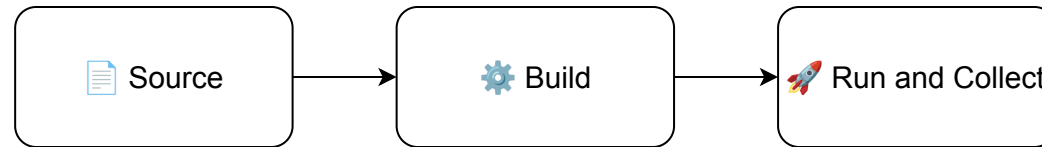


Source

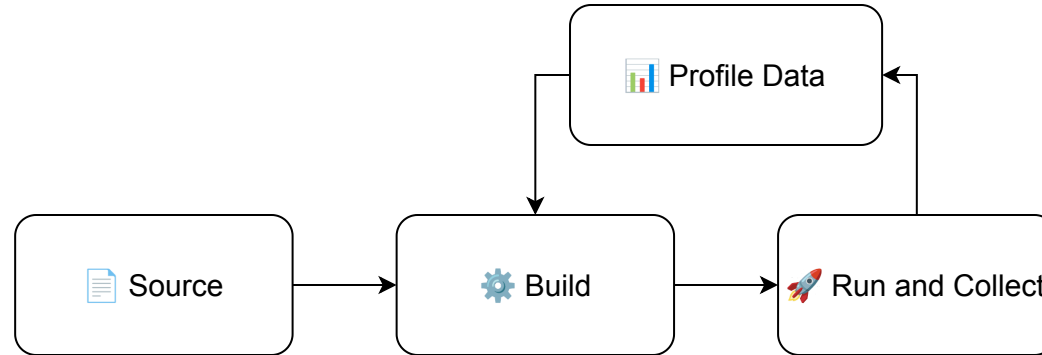
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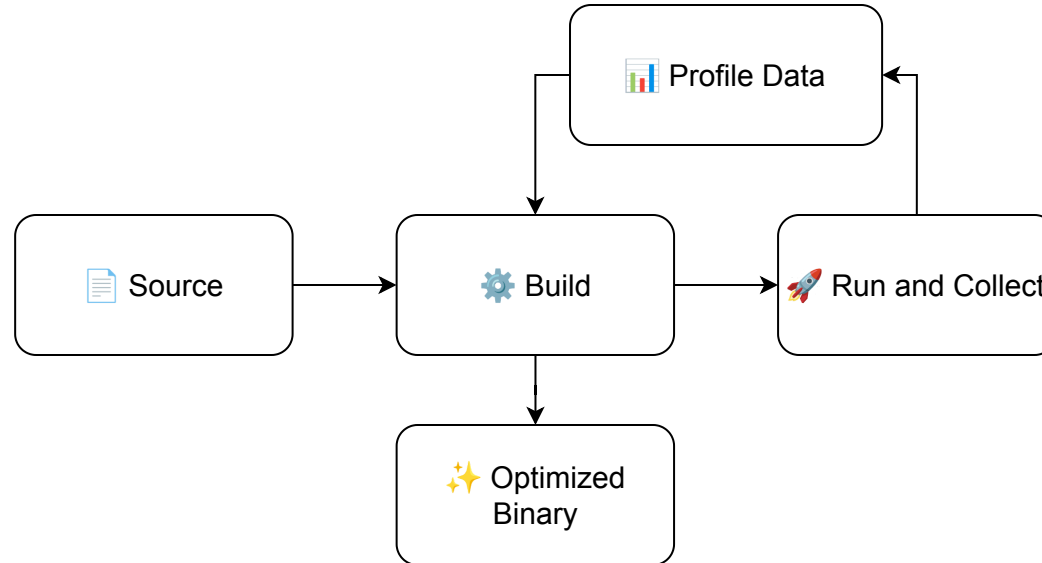
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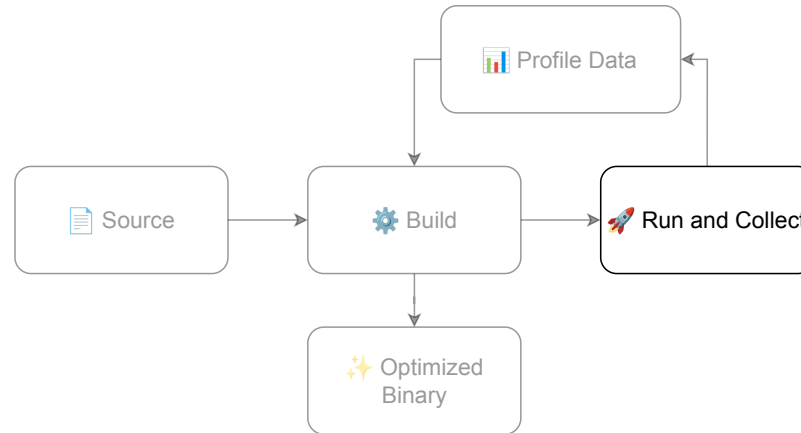
Profile Life-cycle

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- Profiles traverse three main phases within the PGO workflow

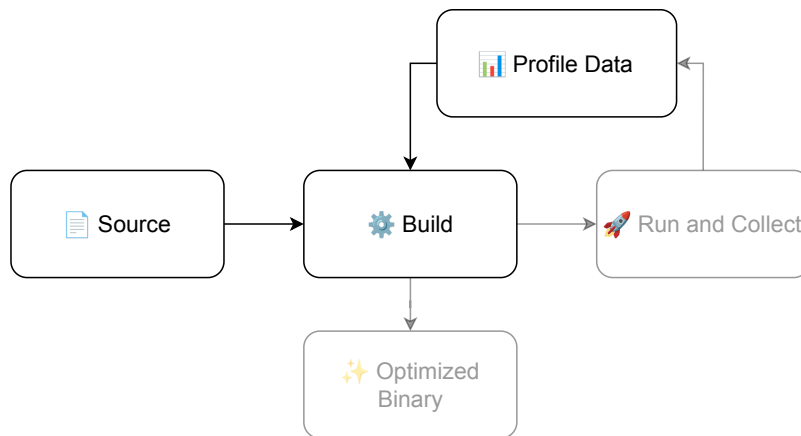
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 - **Collection:** The profile is collected and persisted as a file



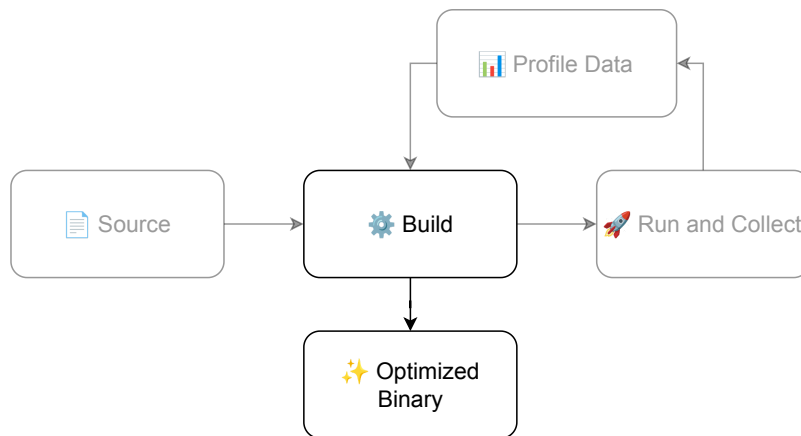
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 - **Collection:** The profile is collected and persisted as a file
 - **Mapping:** The profile is represented as **metadata** attached to program constructs
 - **Usage:** The metadata is used to guide optimizing transformations



Practical Obstacles

- For an ideal PGO application



Profile must be **complete** and **accurate** in each phase of its life-cycle

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- For an ideal PGO application



Profile must be **complete** and **accurate** in each phase of its life-cycle

- Inaccurate profiles may lead the compiler to make suboptimal optimization decisions.
- In practice, each phase hides sources of inaccuracies
 - **Sampling** strategies are inaccurate by nature
 - The dynamic nature of software leads to **stale profiles**
 - Optimizing passes contain errors in **metadata propagation** logic

State of the Art

1. Wenlei He, Julián Mestre, Sergey Pupyrev, Lei Wang, and Hongtao Yu. "Profile inference revisited". ↵
2. Amir Ayupov, Maksim Panchenko, and Sergey Pupyrev. "Stale Profile Matching". ↵
3. Youfeng Wu. "Accuracy of Profile Maintenance in Optimizing Compilers". ↵
4. Profile Information Propagation Unittesting: <https://discourse.llvm.org/t/rfc-profile-information-propagation-unittesting/73595> ↵

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- Lots of effort to solve problems in the collection and mapping phases

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State of the Art

- Lots of effort to solve problems in the collection and mapping phases
 - [1] Proposes a rectification algorithm to rectify sampled profiles
 - [2] Proposes an algorithm to adapt stale profiles to newer program versions
 - [3], [4] only partially address failures in metadata propagation

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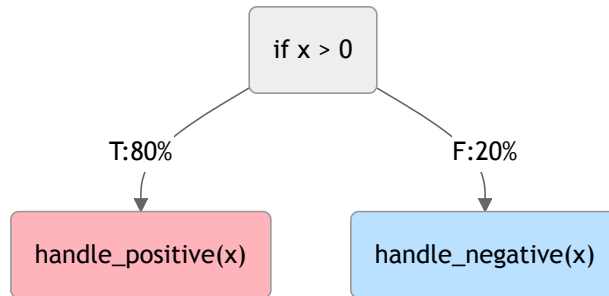
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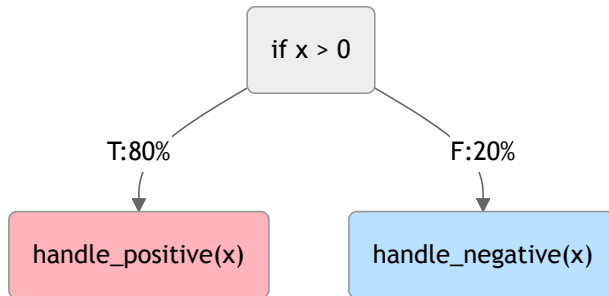
Example of Profile Mishandling

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// Before pass  
if (x > 0) { // Then branch taken 80 times  
    handle_positive(x) 🔥  
} else { // Else branch taken 20 times  
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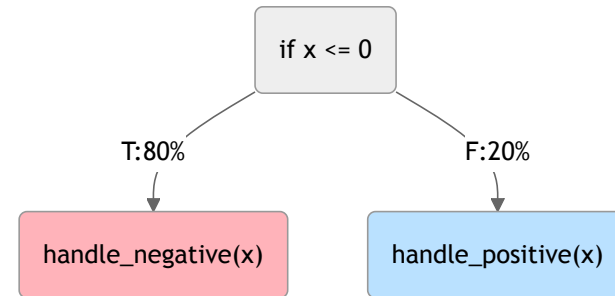


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- **Non-trivial** solution
 - What does it mean to correctly propagate profiles?
 - No immediate correlation between before and after code
 - **Interaction** between passes needs to be taken into account
 - **Static checks** on control-flow are not enough!
 - **No dedicated tools** to validate propagation logic

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Profile information is **transformed** together with the program, but unlike the program itself, its correctness cannot be directly observed.

Research Proposal



Can profile propagation accuracy be assessed systematically?

Research Proposal



Can profile propagation accuracy be assessed systematically?

- A methodological and practical framework to
 - Validate profile information propagation using **random** C programs
 - Improve the PGO logic coverage of the tested compiler via **coverage-guided** testing

Stress-Test approach

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- Stress-testing profile propagation logic with randomly generated programs

Stress-Test approach

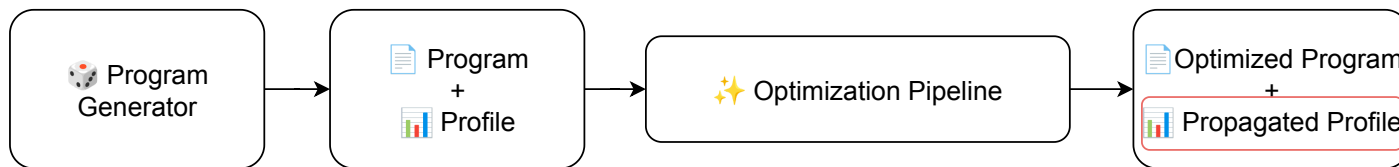
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Program
Generator

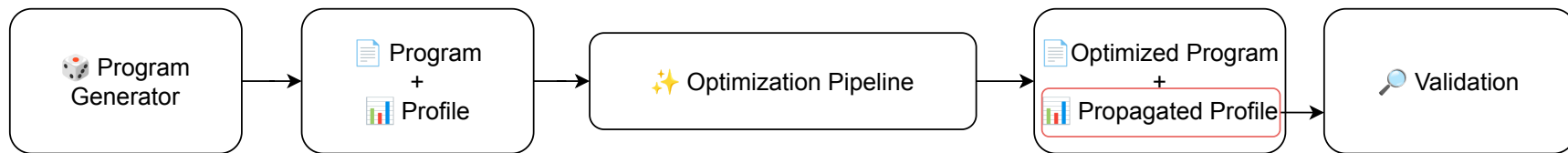
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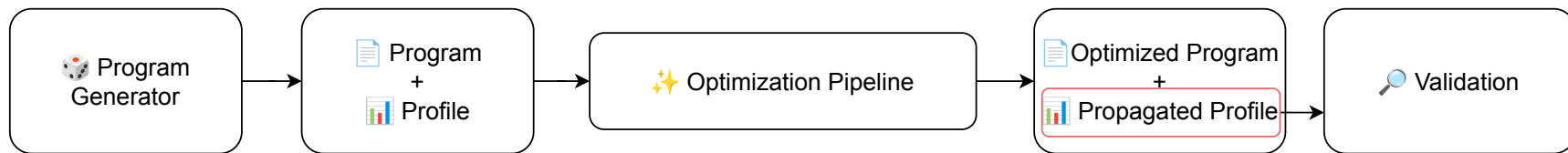
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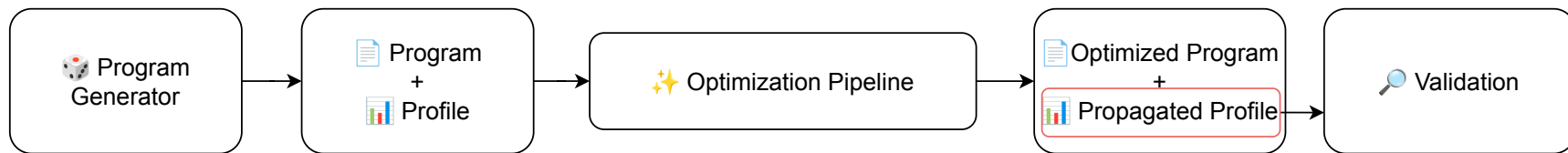
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- Automated **bug triaging** to classify the large number of issues
- Efficacy is bounded by the **complexity** achievable by program generators

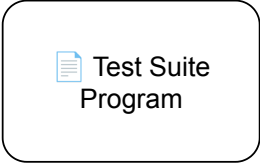
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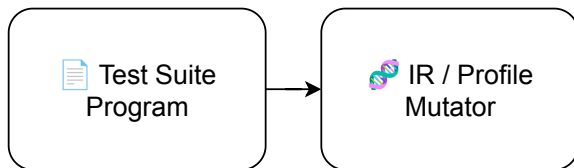
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Test Suite
Program

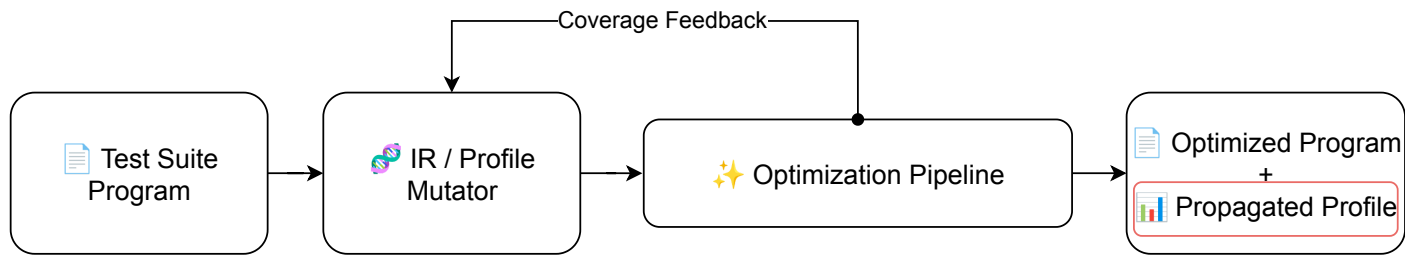
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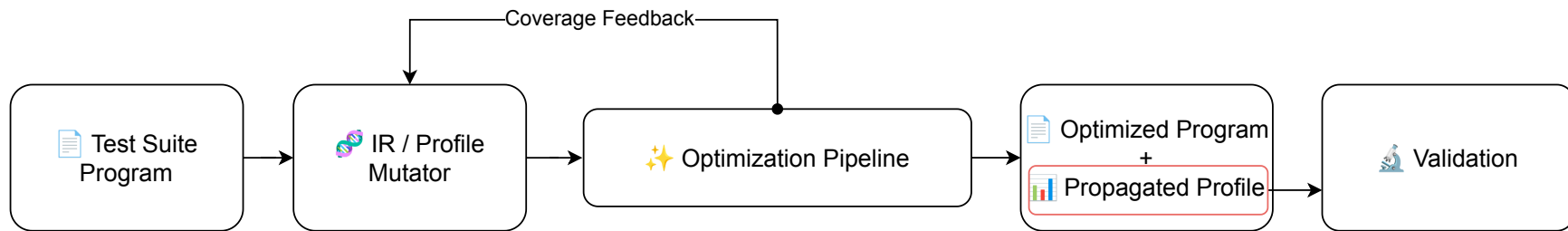
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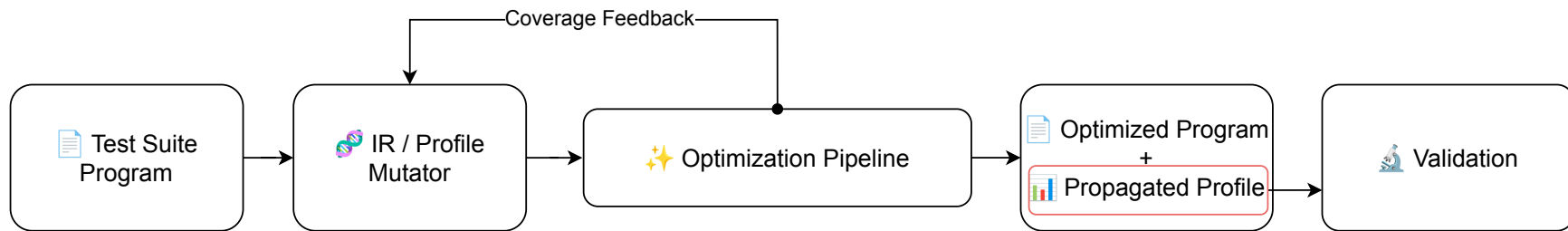
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- Exposes **untested** regions of the compiler to uncover deeper issues

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Bugs



Performance

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- **LLVM** compiler infrastructure as evaluation target
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 - **Improve the performance** of the generated binaries?
 - Improve optimization pass coverage within LLVM?



Bugs



Performance

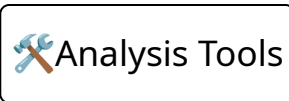


Coverage

Impacts and Benefits

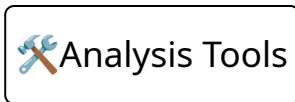
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- **Faster** PGO binaries for the user
- **Energy savings** due to optimal binaries



Analysis Tools



Faster Binaries



Energy Savings

Thanks for the attention!
Questions?